GEOMETRIC VERSION OF THE COHOMOLOGY JUMP LOCI FOR A PROPER FLAT FAMILY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- For a proper morphism of finite presentation with F of finite presentation flat over S, the derived pushforward K = Rf_*F is a perfect complex whose formation commutes with arbitrary base change. invoked at proof lines 8165-8171. Turns the geometric statement into a statement about a perfect complex on S.
- For a perfect complex on S the dimension functions of the cohomology of its derived fibers are upper semicontinuous with constructible level sets compatible with base change. invoked at proof lines 8174-8176. Supplies all three assertions (1), (2), (3) for the fiber cohomology dimensions.

Source reference. Original lemma 0B91: sources/stacks-perfect/pinned-original-source.tex, source lines 7741–7741. Cited at proof lines 8166-8171 to obtain K; the pointwise identification and the application of jump-loci are inside the counted span.

Source reference. Original lemma 0BDI: sources/stacks-perfect/pinned-original-source.tex, source lines 7907–7907. Cited at proof lines 8174-8176 for all three assertions; the reduction of the geometric situation to the perfect complex K is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a proper morphism of finite presentation. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module of finite presentation, flat over S. For fixed $i \in \mathbf{Z}$ consider the function*

$$\beta_i : S \to \{0, 1, 2, \ldots\}, \quad s \longmapsto \dim_{\kappa(s)} H^i(X_s, \mathcal{F}_s)$$

Then we have

(1) *formation of β_i commutes with arbitrary base change,*
(2) *the functions β_i are upper semi-continuous, and*
(3) *the level sets of β_i are locally constructible in S.*

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. By cohomology and base change (more precisely by Lemma 0B91) the object $K = Rf_*\mathcal{F}$ is a perfect object of the derived category of S whose formation commutes with arbitrary base change. In particular we have

$$H^i(X_s, \mathcal{F}_s) = H^i(K \otimes^{\mathbf{L}}_{\mathcal{O}_S} \kappa(s))$$

Thus the lemma follows from Lemma 0BDI. □